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Entanglement assisted probe of the non-Markovian to Markovian transition in open quantum system dynamics

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We utilize a superconducting qubit processor to experimentally probe non-Markovian dynamics of an entangled qubit pair. We prepare an entangled state between two qubits and monitor the evolution of entanglement over time as one of the qubits interacts with a small quantum environment consisting of an auxiliary transmon qubit coupled to its readout cavity. We observe the collapse and revival of the entanglement as a signature of quantum memory effects in the environment. We then engineer the non-Markovianity of the environment by populating its readout cavity with thermal photons to show a transition from non-Markovian to Markovian dynamics, ultimately reaching a regime where the quantum Zeno effect creates a decoherence-free subspace that effectively stabilizes the entanglement between the qubits.

Decoherence is a ubiquitous challenge in quantum technologies. At a microscopic level, decoherence arises from the entanglement of a quantum system with degrees of freedom in its environment. Without access to these degrees of freedom, information about the quantum state is lost [1, 2]. The monotonic reduction in a quantum state's coherence is typically described by the well-known Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) master equation [3, 4] for the system's density operator ρ. In particular, the GKSL master equation is valid under the Born-Markov set of approximations, which assume both weak coupling to the environment, and that the environment is Markovian, i.e. memoryless [5]. This mathematically amenable description is surprisingly effective in describing a broad range of quantum dynamics. Moreover, in the Markovian regime, dissipation engineering by an intentional introduction of Markovian dissipation has been employed as a powerful method of quantum control; with applications including error correction [6–8], state preparation [9, 10], state stabilization [11, 12], and quantum simulation [13]. Naturally, however, there is another paradigm of decoherence known as the non-Markovian regime, where quantum memory effects induced by large system-environment correlations thwart a Markovian description. In this regime, the dynamics of the system is governed by the generalized Nakajima-Zwanzig master equation [14, 15] which incorporates the memory effects of the environment.

Non-Markovian dynamics have the potential to enable novel applications stemming from memory effects in the environment, such as new approaches towards fault-tolerant quantum computation [16–18], quantum control [19], fidelity improvement in the implementation of the teleportation algorithms [20], and coherence preservation [21]. The non-Markovianity of an open quantum system can be measured using two common methods [22]. The most prominent measure is known as the *trace distance* method proposed by Breuer et. al. [23], which relies on the fact that any completely positive trace-preserving (CPTP) quantum map between two-time steps will only result in a decrease of the distinguishability between two quantum states, hence any increase in the distance between states is associated with memory effects [24]. Later, an *entanglement measure* was introduced by Rivas et. al. [25], where one probes quantum memory effects by allowing part of an entangled pair to interact with an environment. Again, a CPTP map will only decrease the degree of entanglement and an increase in entanglement during the system evolution is a signature of quantum memory effects. Both methods [26] have been employed to observe signatures of non-Markovianity, notably in nitrogen-vacancy centers [27–30], photonic systems [31–35], nuclear magnetic resonance [35–37], trapped ions [38], and on superconducting processors [39].

In this Letter, we harness the entanglement between two superconducting qubits as a probe of quantum memory effects. We initialize the qubits in a Bell state and study the qubits' concurrence [40] over time as one of the qubits interacts with a small quantum environment consisting of a third qubit dispersively coupled to a microwave resonator. We observe collapse and then revival of the qubits' concurrence as a clear signature of the non-Markovian nature of the environment as the qubit becomes entangled and then disentangled with the en-

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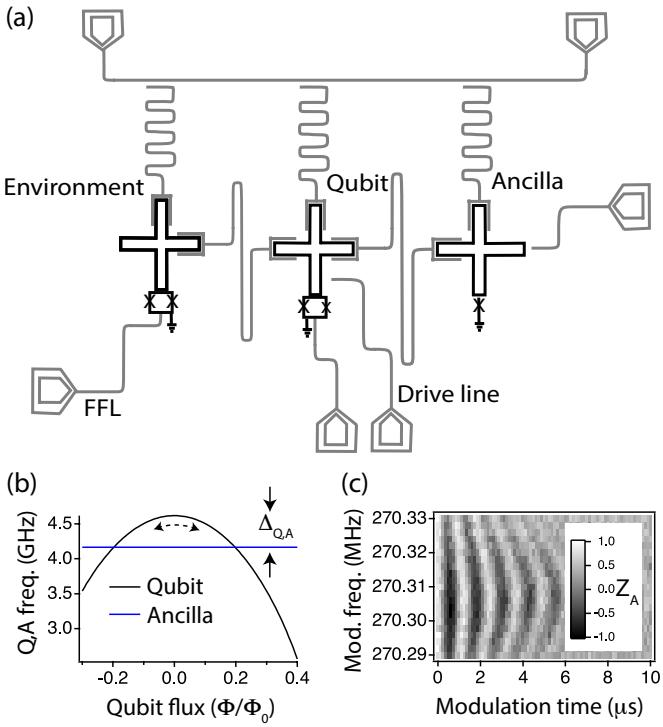

FIG. 1. **Experiment setup.** (a) Sketch of the experiment which includes three qubits respectively labeled "Environment", "Qubit", "Ancilla". The qubits share resonators that mediate nearest-neighbor coupling. Each qubit is coupled to a readout resonator, which can be probed by a common feedline. The Environment and Qubit are frequency tunable via on-chip fast flux lines (FFL). (b) The respective frequencies of the Qubit and Ancilla; resonant coupling between the qubits is achieved by applying a parametric modulation of the Qubit at roughly $\Delta_{\mathrm{Q,A}}/2$. (c) When the Qubit is prepared in its excited state, parametric resonance can be observed by examining the Ancilla excitation versus modulation frequency.

vironment. The non-Markovianity of the environment is then tuned by introducing Lindblad dephasing on the environment [41]. This allows us to investigate a transition away from non-Markovian dynamics to a regime where the GKSL master equation describes the dynamics. Since the GKSL master equation requires a Markovian approximation, we refer to this regime as the Markovian regime. In this Markovian regime, we further increase the dissipation on the environment, ultimately reaching a regime where the quantum Zeno effect pins the environment state, thereby preserving the qubits' entanglement.

Figure 1(a) displays the basic setup of the experiment, with the system Hamiltonian given in [42]. The experiment comprises a three-qubit processor with individual readout resonators dispersively coupled to each qubit and nearest-neighbor qubits sharing a resonator mediated coupling. The readout resonators allow us to perform individual state readouts of the three qubits by probing the associated microwave resonators with a microwave drive. We first focus on a sub-portion of the processor with two qubits denoted as the "Qubit" and the "Ancilla". The Qubit is frequency tunable via a Superconducting QUantum Interference Device (SQUID) loop and the Ancilla is fixed-frequency, both designed to be in the transmon regime [43]. In order to minimize the decoherence effects from flux noise, we operate the Qubit at its flux sweet spot and introduce coupling to the Ancilla via parametric modulation [44]. To this end, we apply an ac radio frequency drive on the Qubit fast flux line at roughly half the detuning between the Qubit and Ancilla (Fig. 1(b)). We identify the resonance condition between the Qubit and Ancilla by initializing the qubit in its excited state and then applying the parametric modulation for a variable duration. Figure 1(c) shows the time evolution of the Ancilla $\langle\sigma_z\rangle \equiv Z_A$ near the parametric resonance. We observe a clear chevron profile with detuning from which we extract a parametric coupling rate of $\Omega_{\mathrm{Q,A}}/2\pi = 0.477$ MHz.

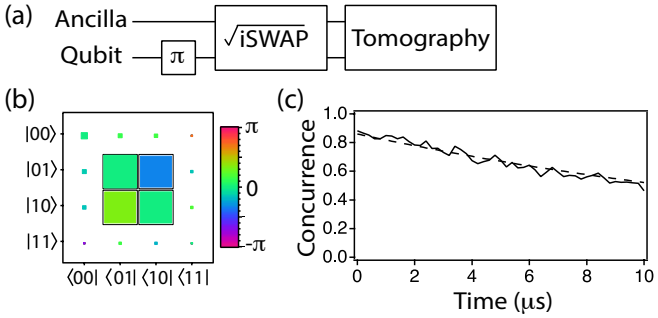

FIG. 2. **Qubit-Ancilla entanglement.** (a) We prepare an entangled state by initializing the Qubit in the excited state, and then applying a $\sqrt{i\mathrm{SWAP}}$ gate via parametric modulation. (b) Quantum state tomography allows us to reconstruct the Qubit–Ancilla density operator, yielding an entangled state of the form $\frac{1}{\sqrt{2}}(|10\rangle + e^{i\phi}|01\rangle)$. (c) The measured Qubit–Ancilla concurrence versus time (solid line); we observe a monotonic decrease in the entanglement over time consistent with the single-qubit dephasing rates (dashed line).

We utilize this parametric coupling to produce a Bell state between the Qubit and Ancilla, as depicted in Fig. 2(a). After applying a π rotation to the Qubit, we activate the parametric coupling for 530 ns, corresponding to a $\sqrt{i\mathrm{SWAP}}$ gate, in principle, leaving the Qubit and Ancilla in a state, $\frac{1}{\sqrt{2}}(|10\rangle + e^{i\phi}|01\rangle)$ [45]. We utilize quantum state tomography of the Qubit and Ancilla to characterize the resulting entangled state. For this, we measure 9 Pauli expectation value pairs, $\{\langle\Sigma_{\mathrm{Q}}\Sigma_{\mathrm{A}}\rangle\}$, with $\Sigma_{\mathrm{Q,A}} \in \{X, Y, Z\}$ by simultaneously measuring the state of both the Qubit and Ancilla [46]. The average readout fidelities of the Qubit and Ancilla are respectively 0.97 and 0.96. As discussed in the Supplementary Materials, we use a maximum likelihood estimation method [47] to determine the components of the Qubit–Ancilla density matrix, displayed in Fig. 2(b). We observe a Bell state fidelity of 0.91, corresponding to a concurrence of 0.89.

With the Qubit and Ancilla entangled, we now study

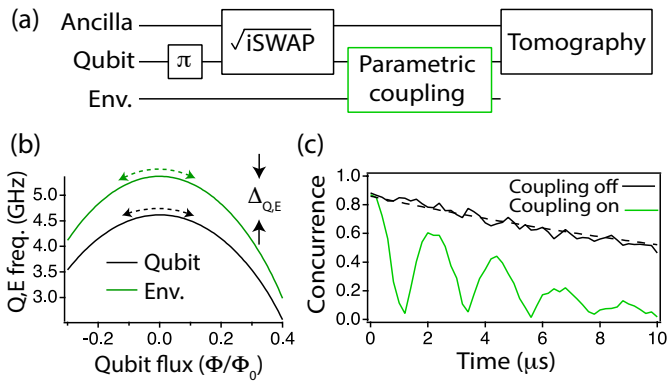

FIG. 3. **Concurrence revival due to non-Markovianity.** (a) We prepare an entangled state between the Qubit and Ancilla, then we apply a parametric coupling between the Qubit and Environment, finally, we perform a set of tomography pulses to reconstruct the density matrix of the Qubit–Ancilla subspace. The concurrence evolution is realized by varying the length of the Qubit–Environment parametric coupling pulses. (b) The respective frequencies of Qubit and Environment showing the Qubit–Environment detuning. (c) Concurrence evolution as a function of the Qubit–Environment parametric coupling pulse length (green). The black curve shows the concurrence evolution when the parametric coupling is turned off.

the evolution of the entanglement over time as the system sits idle. We display the Qubit–Ancilla concurrence versus time in Fig. 2(c). The concurrence slowly decreases over a timescale consistent with the respective individual dephasing times of the Qubit ($T_2^{*(Q)} = 39\ \mu s$) and Ancilla ($T_2^{*(A)} = 41\ \mu s$), e.g. $\mathcal{C} \propto \exp(-t/T_2^{*(Q)} - t/T_2^{*(A)})$, as given by the dashed line in Fig. 2(c).

We now turn to studying the interaction of the Qubit–Ancilla subspace with the Environment. As displayed in Fig. 3(a), after preparing the Qubit–Ancilla in an initial Bell state, we introduce a parametric coupling between the Qubit and Environment. In this case, we apply flux modulation simultaneously to both the Qubit and Environment (Fig. 3(b)) bringing the two into parametric resonance. Both Qubit and Environment are modulated at approximately one-quarter of their detuning ($\Delta_{\mathrm{Q,E}}/4 = 2\pi \times 175$ rad.$/\mu$s), which introduces a resonant transverse coupling between the Qubit–Environment pair at a rate of $\Omega_{\mathrm{Q,E}} = 2\pi \times 0.473$ rad.$/\mu$s, limited by the resonator-mediated coupling between the pair. After applying the parametric coupling between Qubit and Environment, we perform quantum state tomography on the Qubit–Ancilla subsystem to determine the remaining concurrence.

Figure 3(c) displays the evolution of the concurrence when the interaction between the Qubit and Environment is introduced. In comparison to the monotonic decrease in entanglement observed previously (black curve), we now note a rapid decrease in entanglement, with clear revivals at later times (green curve). The initial decrease is expected from the principle of monogamy of entanglement [48]. Since the Qubit–Ancilla system is in a maximally entangled state, the entanglement between the Qubit–Environment introduced by the parametric coupling must cause the Qubit–Ancilla entanglement to decrease. The revival of entanglement occurs as the Qubit–Environment coupling continues and the Environment state is swapped back into the Qubit. This revival of entanglement is a clear indicator of non-Markovianity, indicating that the environment has quantum coherent memory. This is indeed expected since the environment is itself a simple two-level system. The non-Markovianity of the system can be can be calculated as [25],

$$\mathcal{N} = \int_{t_0}^{t_f} dt \left| \frac{d\mathcal{C}[\rho_{\mathrm{Q,A}}(t)]}{dt} \right| - \Delta\mathcal{C}, \tag{1}$$

where $\mathcal{C}[\cdot]$ denotes the concurrence measure, $\Delta\mathcal{C}$ is the difference in the concurrence at the initial and final steps of the evolution, and ρ_{QA} represents the Qubit–Ancilla density matrix. To elaborate, we look at the time derivative of the concurrence over the entire time evolution of the system $\in [t_0 = 0\ \mu s, t_f = 10\ \mu s]$ at discrete time steps. It is clear from Eq. 1 that the positive slope of the concurrence contributes to the non-Markovianity measure. By applying Eq. 1 to the data in Fig. 3(c), we achieve a non-Markovianity of $\mathcal{N} = 1.4$.

With a clear demonstration of non-Markovian dynamics, we now study how this measure changes as the memory of the Environment is tuned. We achieve this by expanding the size of the Environment to include the quantum states of light that occupy the microwave resonator that is dispersively coupled to the Environment. So far, we have considered this resonator to remain in the vacuum state, which does not affect the Environment's memory. Now, we introduce pseudo-thermal photons into this resonator via a noisy microwave drive as indicated in Fig. 4(a). The interaction between the Environment and its resonator is captured by the simple dispersive coupling Hamiltonian, $H_{\mathrm{int}} = \chi a^\dagger a \sigma_z^{\mathrm{E}}$, where $\chi/2\pi = 200$ kHz is the dispersive coupling rate, $a^\dagger a$ is the resonator photon number, and σ_z^{E} is the Pauli operator that acts on the Environment in the energy basis. This interaction can be viewed as either an Environment-state-dependent frequency shift on the resonator frequency, whereby photons carry away information about the state of the Environment, or as an ac-Stark shift of the qubit frequency, whereby the fluctuating intra-resonator photon number dephases the qubit [9, 49].

The noisy drive on the cavity is chosen to have a bandwidth (1.8 MHz) that exceeds χ, ensuring a uniform drive independent of the Environment state. Furthermore, this drive is set to have a correlation time (90 ns) much shorter than any other timescale of the dynamics, allowing us to treat its dephasing effect as Markovian. We calibrate the dephasing via direct Ramsey measurements on the Environment. This establishes a relationship between the dephasing rate and the noise amplitude (A_{in}) as shown in Fig. 4(b). We find an empirical relationship

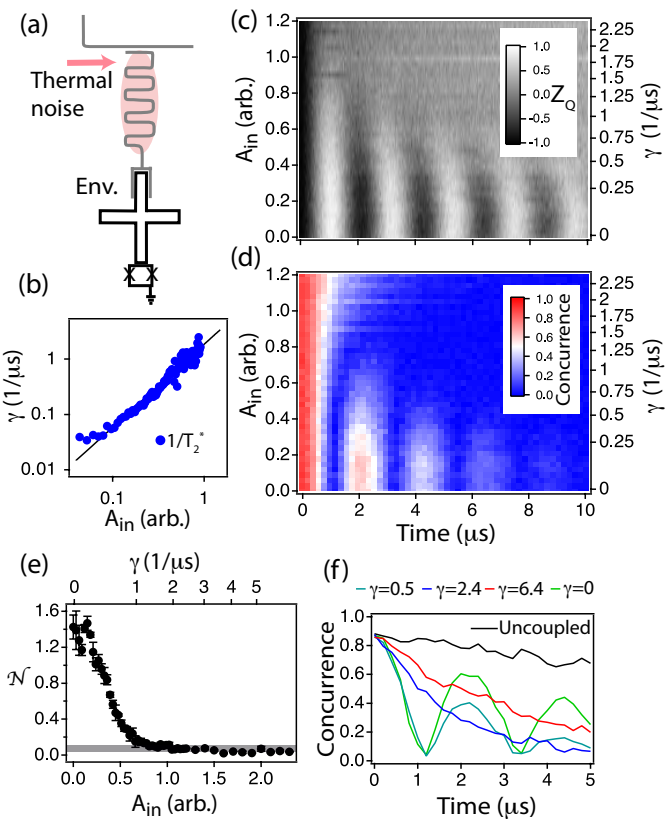

FIG. 4. **Non-Markovian to Markovian transition.** (a) By driving the Environment's readout resonator with pseudo-thermal noise of amplitude $A_{\rm in}$ we tune the Environment's memory. (b) This memory is quantified through Ramsey measurements on the Environment to determine the dephasing rate γ versus $A_{\rm in}$. (c) For each value of $A_{\rm in}$ we calibrate the frequency of the parametric drive between the Qubit and Environment by studying Z_Q versus time and maximizing the population transfer [42]. (d) The Qubit–Ancilla concurrence versus time for different Environment dephasing rates. The transition to monotonic behavior indicates the transition from non-Markovian to Markovian dynamics. (e) The non-Markovian measure (1) quantified across the transition. The error bars indicate the standard error of the mean from three independent experimental trials. The gray bar indicates the measure applied to the case where the Environment is decoupled and characterizes the background of the measure. (f) The concurrence versus time for a few specific dephasing rates (expressed in units of μs^{-1}).

for the Environment dephasing $\gamma = 1.84(\mu\text{s})^{-1}A_{\rm in}^{1.5}$, as given by the black line in Fig. 4(b).

The introduction of the thermal photons into the Environment causes slight shifts in the parametric coupling between the Qubit and the Environment. As such, we calibrate the parametric coupling between the Qubit and Environment for each value of $A_{\rm in}$. Figure 4(c) shows the resulting parametric coupling between the Qubit and Environment when the Qubit is initialized in the exited state and the parametric coupling is activated for a variable duration of time. By increasing the dephasing of the

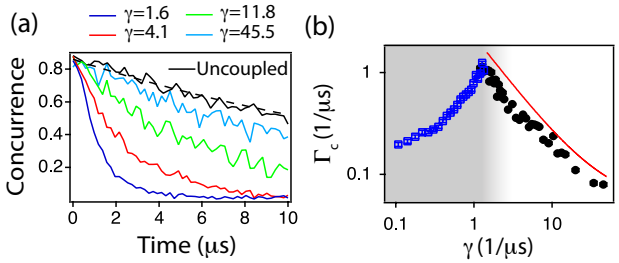

FIG. 5. **Quantum Zeno stabilization of entanglement.** (a) Qubit–Ancilla concurrence versus time for different Environment dephasing rates; as the dephasing increases, the entanglement decay approaches the uncoupled case consistent with the Qubit and Ancilla's individual dephasing rates. (b) The exponential decay rate of the concurrence ($\Gamma_{\rm c}$) versus Environment dephasing rate. The gray region indicates the non-Markovian regime, where we determine $\Gamma_{\rm c}$ by fitting the overall (non-monotonic) decay envelope of the concurrence. In the Markovian regime, we observe that the Zeno effect suppresses the concurrence decay induced by the environment, in agreement with the expected scaling (red line).

Environment, we observe diminished population transfer contrast between the Qubit and the Environment.

Next, we investigate the time evolution of the Qubit–Ancilla concurrence for different values of the Environment dephasing. Increasing the Environment dephasing induces a transition from non-Markovian to Markovian dynamics as displayed in Fig. 4(d). We quantify the transition away from non-Markovian dynamics via the measure (Eq. 1) as displayed in Fig. 4(e); as the dephasing of the Environment is increased beyond $\gamma \simeq 1$ $(\mu\text{s})^{-1}$, $\mathcal{N}$ becomes consistent with zero. However, the dynamics are not immediately Markovian, which we define by the applicability of the GKSL master equation to the Qubit–Ancilla subsystem. As we study in [42], the GKSL master equation yields exponentially decaying dynamics of the concurrence. This matches well the measured dynamics for $\gamma \gtrsim 3$ $(\mu\text{s})^{-1}$, but fails to capture the dynamics for smaller values of γ. As such, the transition between these two regions, as defined, is not abrupt.

In Fig. 4(f) we display the concurrence versus time for a few selected values of γ. We note two important trends; first, in the non-Markovian regime, increasing dephasing accelerates the decay envelope of the concurrence (compare $\gamma = 0$ and $\gamma = 0.5$), and second, in the Markovian regime, further increasing dephasing slows the decay of the concurrence ($\gamma = 2.4$ and $\gamma = 6.4$). This can be understood within the context of the quantum Zeno effect [50–57]. The thermal photons perform measurement (at rate γ) of the Environment, which slows the coupling induced by the parametric drive. In Fig. 5, we explore in detail how the dephasing of the Environment affects the Qubit–Ancilla entanglement. Figure 5a displays the concurrence versus time for several values of the dephasing in the Markovian regime. For increasing measurement on the Environment, we see that the decay of concurrence is

slowed, approaching the limiting case where the Qubit is completely uncoupled from the Environment. We characterize the exponential decay of the concurrence with a rate Γ_c, and display this rate versus Environment dephasing in Fig. 5b in both the non-Markovian and Markovian regimes; the transition between these two regimes coincides with the onset of Zeno stabilization of the entanglement. Under the standard analysis of the Zeno effect [55, 58], we expect $\Gamma_c = \Omega^2_{Q,E}/4\gamma + \Gamma_0$. Here $\Gamma_0 = 1/T_2^{*Q} + 1/T_2^{*A}$ is the decay rate of the concurrence when the Environment is decoupled. We observe close agreement with this expected scaling (red curve). This demonstrates a new approach to preserving quantum entanglement via Zeno-enabled pinning of environment states.

In conclusion, we have quantified the transition from non-Markovian dynamics to Zeno dynamics with an entanglement-assisted probe. Importantly, the probe is sensitive to the *quantum memory* of the environment; a classical environment that stores populations will not result in the revival of concurrence for the entangled probe. This approach can have utility in the test of the quantum nature of decoherence channels (e.g. in testing models of quantum gravity [59]). Moreover, by introducing controllable dissipation on the environment we observe stabilization of the Qubit–Ancilla subsystem, highlighting how dissipation forms a powerful tool for quantum subspace engineering [13].

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